(https://ieeexplore.ieee.org/document/1353004). Detecting failures is a fundamental concern for fault tolerance in distributed systems.

The ϕ accrual failure detector calculates a value on a continuous scale (called a ϕ [phi] value) instead of determining a Boolean value indicating failure (if the node is reachable or not). From the referenced paper: “Roughly speaking, this value captures the degree of confidence that a corresponding monitored process has crashed. If the process crashes, the value is guaranteed to accrue over time and tend toward infinity, hence the name.” This value is used as an indicator for suspecting that something is wrong (a suspicion level) instead of determining a hard-and-fast yes-or-no result.

The suspicion level concept makes the failure detector tunable and allows for a decoupling between application requirements and monitoring the environment. The cluster module provides settings for the failure detector that you can tune for your specific network environment in the `akka.cluster.failure-detector` section, including a threshold for the ϕ value at which a node is deemed unreachable.

Nodes are often deemed unreachable when they’re in a *GC pause* state, which means it’s taking far too long to finish garbage collection, and a JVM can’t do anything else until garbage collection has completed.

In this example, you looked at the cluster logs to determine the cluster’s state and see how to interact with it. You entered commands into the console to get nodes to leave the cluster or switch to the `Down` state. As mentioned earlier, this is not the approach you use in a real application—all these tasks are delegated to the SBR module.

A special module is available to interact with the cluster, check its status, and change it: Akka Management, particularly the Cluster HTTP submodule. Let’s take a look at it.

8.3 Akka Management and the Cluster HTTP extension